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10-bit 50MS/s SAR-ADC

HARSHIT GUPTA

UIN: 735006036

TEXAS A&M UNIVERSITY
College Name Goes Here

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- System Architecture
- Block-Level Designs (Switch, Comparator, CDAC, Digital-logic)
- Testbench and Sim Results
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SYSTEM ARCHITECTURE

- Monotonic Switching (helps in saving power)
- Asynchronous SAR Logic (only one clock is required)
- Split C-DAC (Helps to reduce the acquisition time without impacting the SNDR) as this schemes reduces the effective cap experienced by the S/H switch

- Ts = 1/(50M) = 20ns
- Acquisition Time = 5ns, Conversion Time=15ns (25%-75%, not suitable for Time-interleaved ADC. This was made 50-50 over there.)
- 7RC << 5ns , R~3ohms, Ceq = 32pF (in the split-CDAC scheme)

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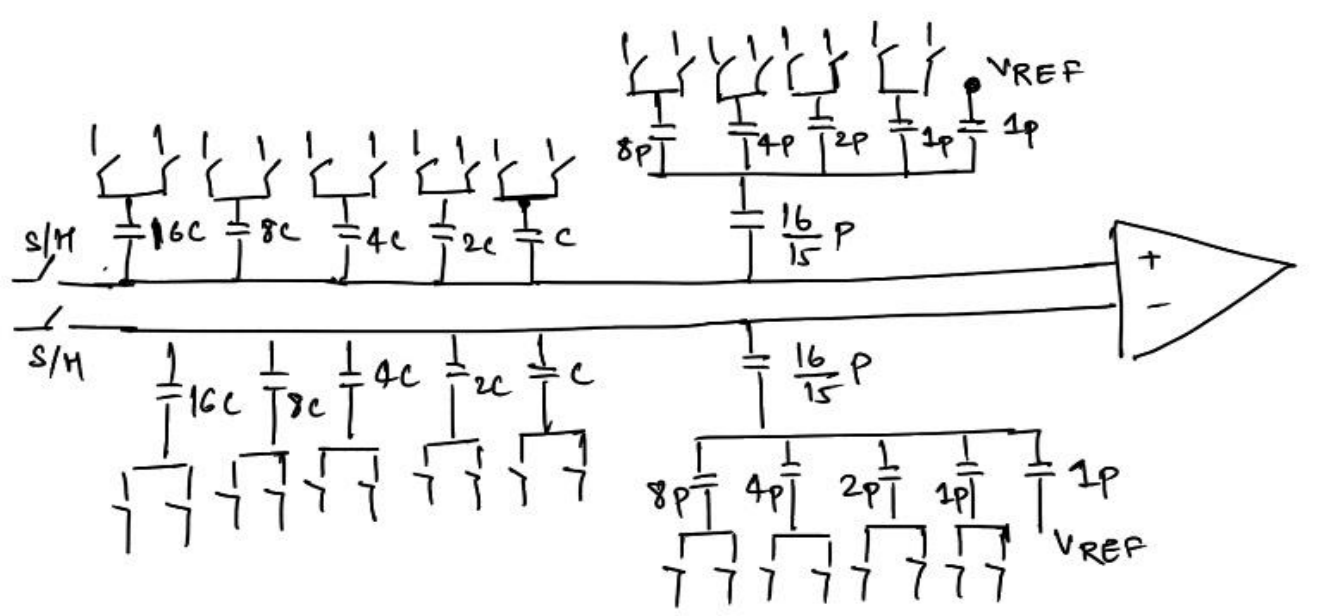
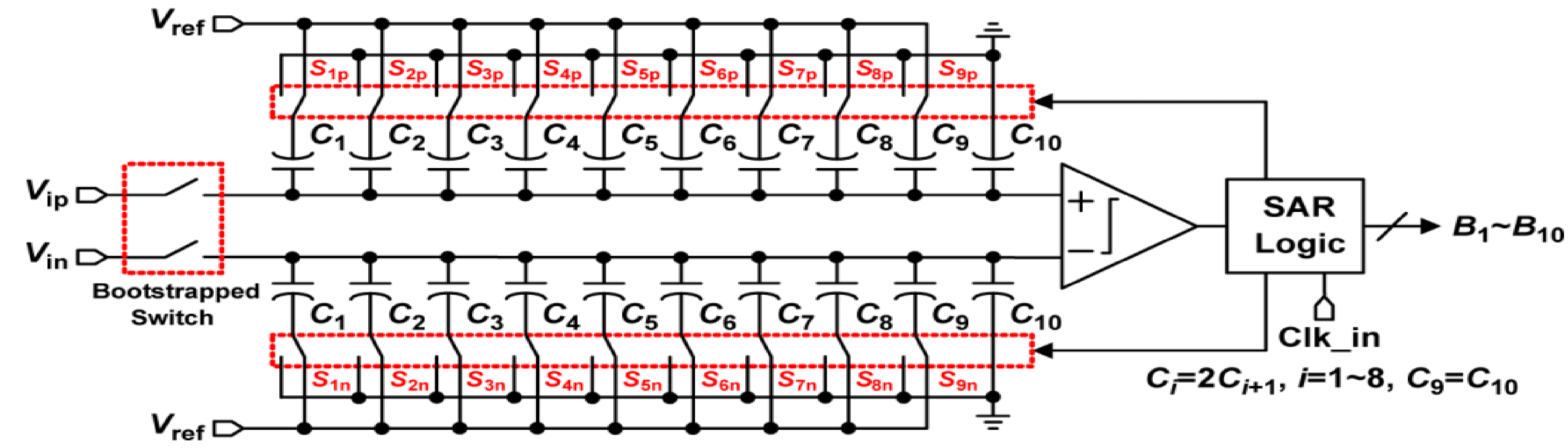
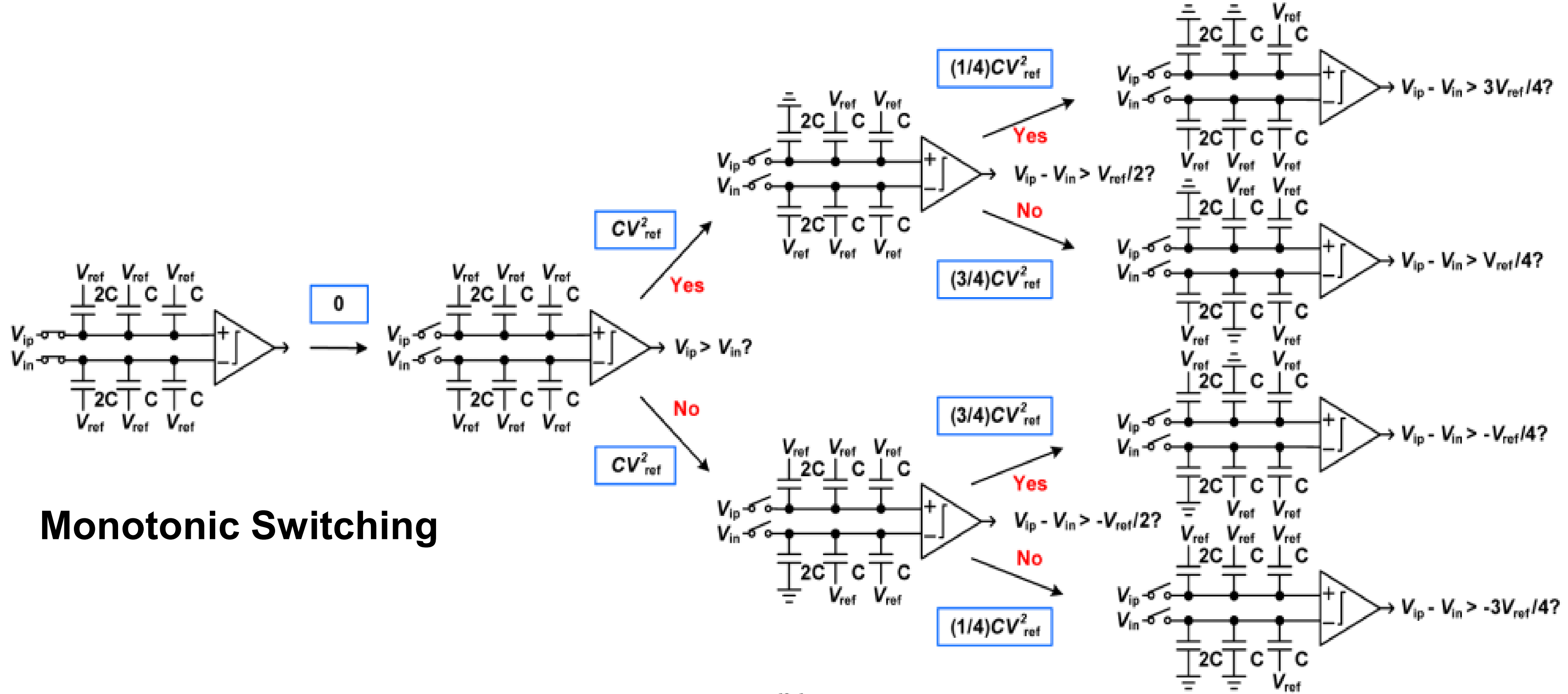


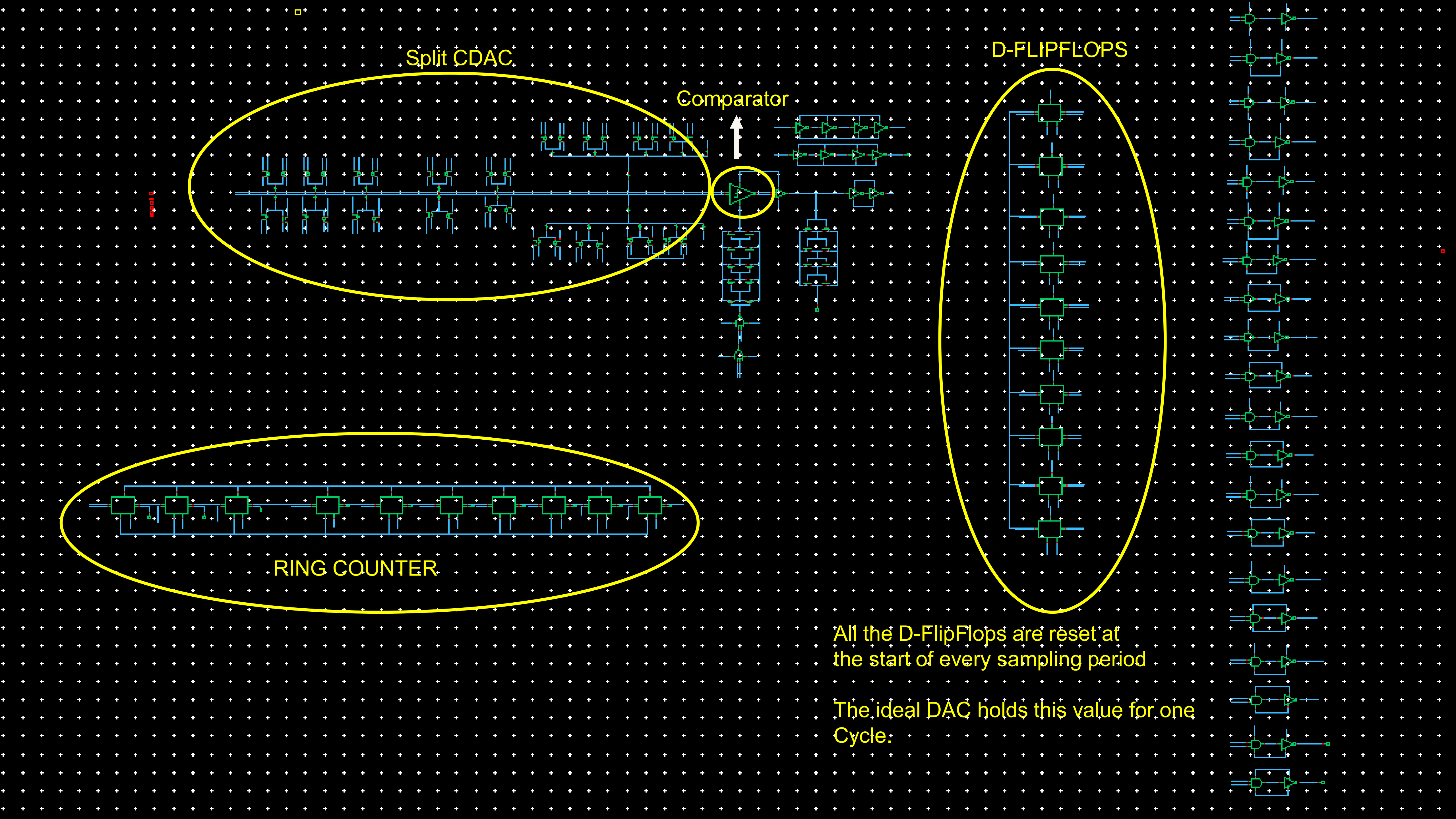
Key Issue with Split CDAC: Realizing the bridge capacitor which Is not integer multiple of the Cmin (1pF).

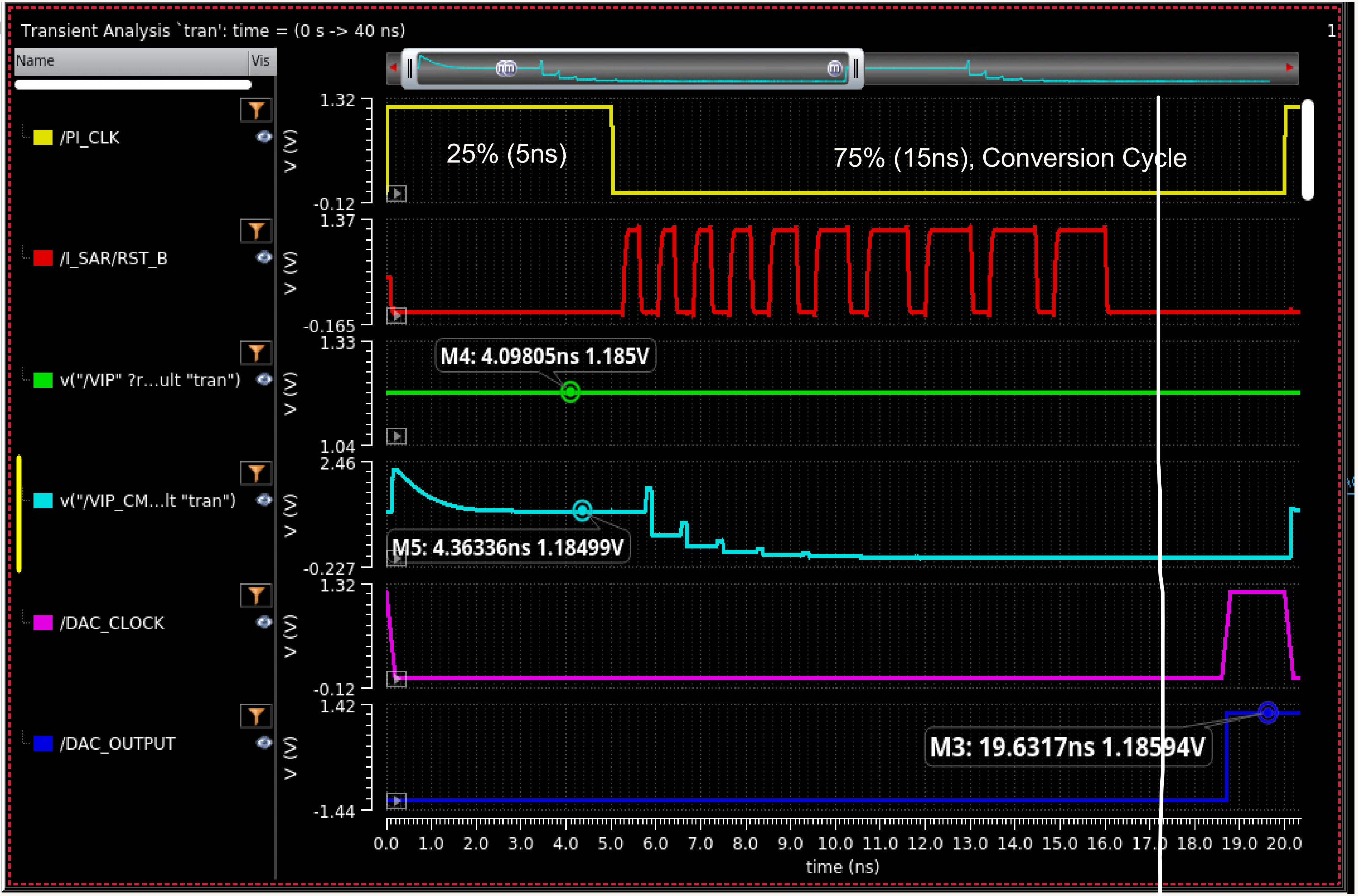


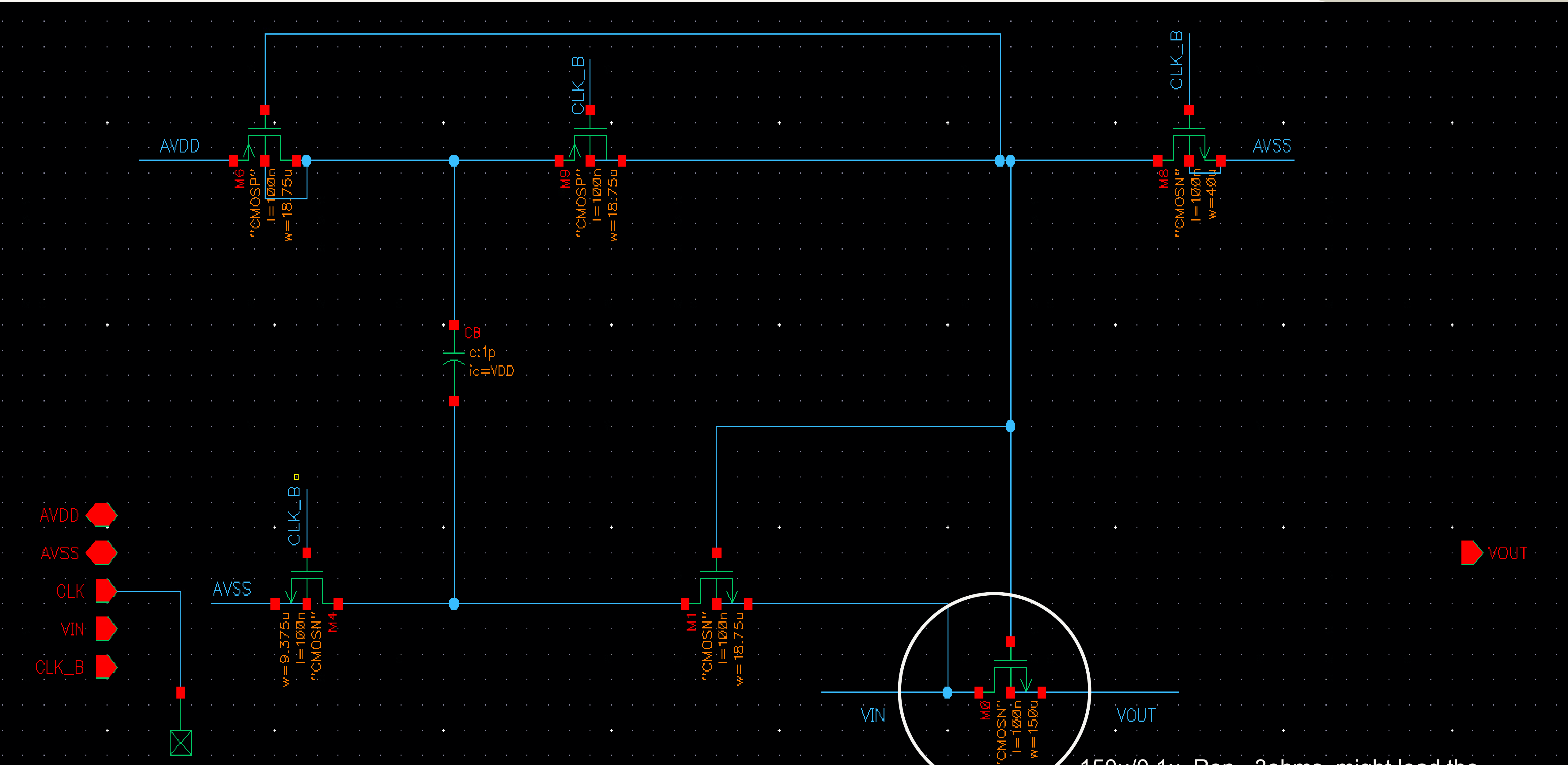


Monotonic Switching

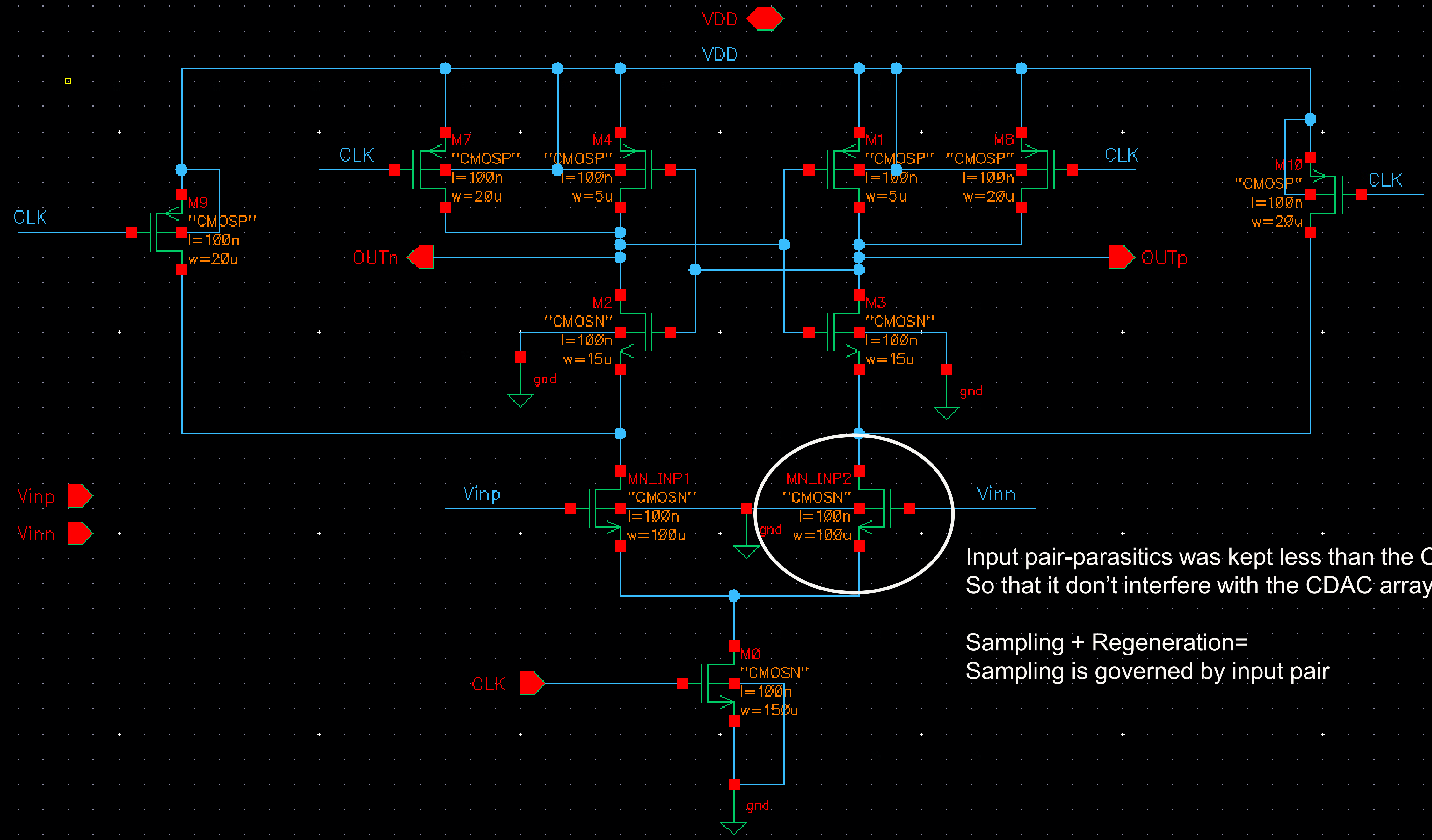
(b)


Split CDAC
Comparator
D-FLIPFLOPS
RING COUNTER
All the D-FlipFlops are reset at
the start of every sampling period
The ideal DAC holds this value for one
Cycle.


Transient Analysis `tran': time = (0 s -> 40 ns)
Name
Vis
/PI_CLK
/I_SAR/RST_B
v("/VIP" ?r...ult "tran")
v("/VIP_CM...lt "tran")
/DAC_CLOCK
/DAC_OUTPUT
25% (5ns)
75% (15ns), Conversion Cycle
M4: 4.09805ns 1.185V
M5: 4.36336ns 1.18499V
M3: 19.6317ns 1.18594V
V (V)
1.32
-0.12
1.37
-0.165
1.33
1.04
2.46
-0.227
1.32
-0.12
1.42
-1.44
0.0 1.0 2.0 3.0 4.0 5.0 6.0 7.0 8.0 9.0 10.0 11.0 12.0 13.0 14.0 15.0 16.0 17.0 18.0 19.0 20.0
time (ns)



150u/0.1u, Ron~ 3ohms, might load the ADC pre-driver buffers.

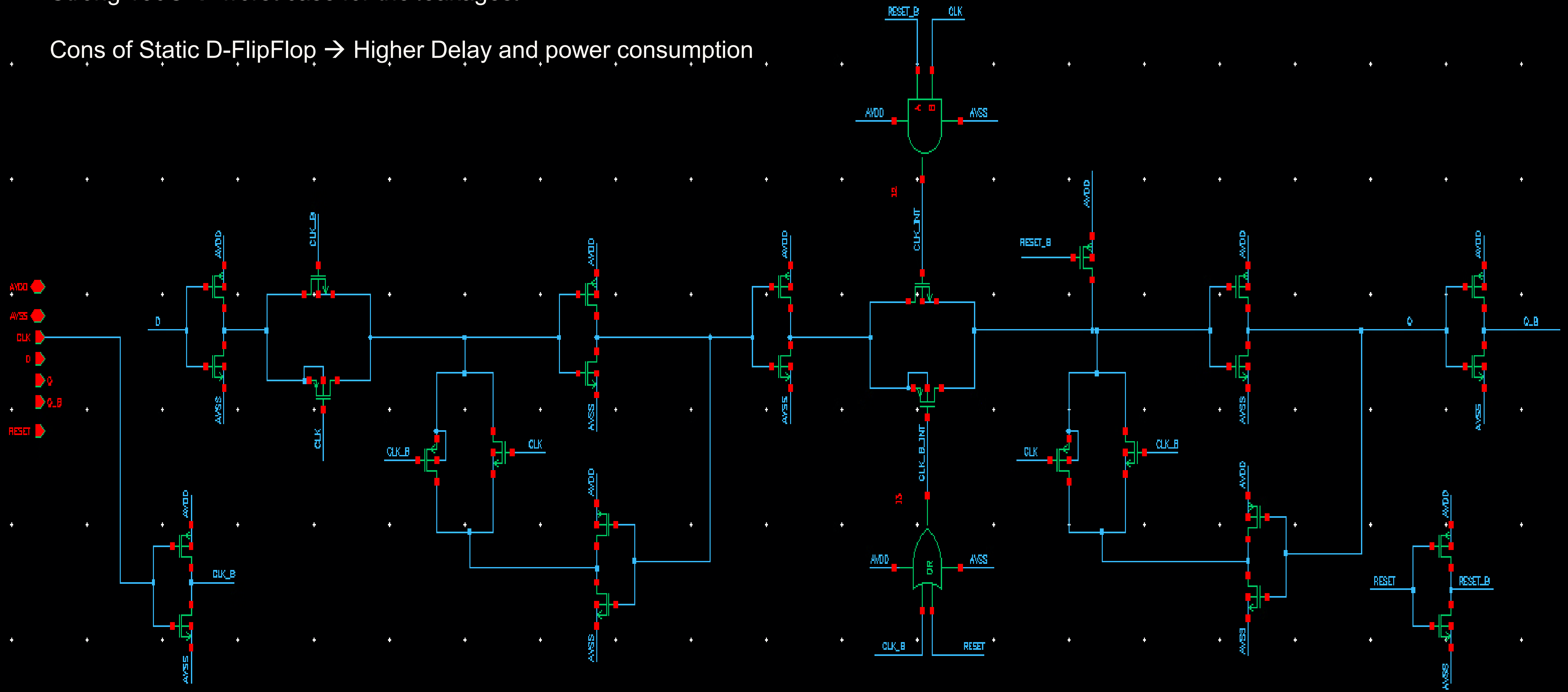


Input pair-parasitics was kept less than the Cmin
So that it don’t interfere with the CDAC array.

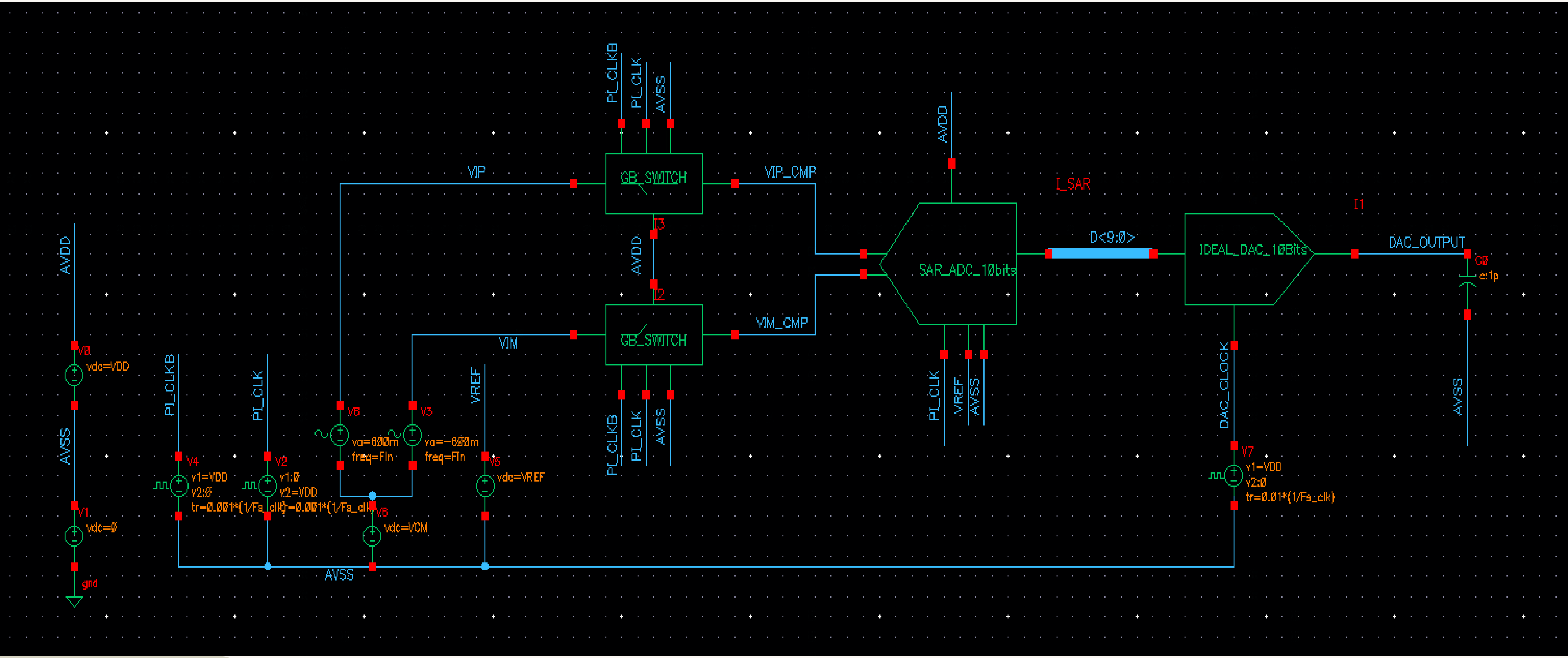
Sampling + Regeneration=
Sampling is governed by input pair

Static D-FlipFlop → slower but immune to leakages (which might be a issue with Dynamic D-FlipFlops for MHz clocks)
Strong 150C → worst case for the leakages.

Cons of Static D-FlipFlop → Higher Delay and power consumption



TESTBENCH AND SIMULATION RESULTS



Sine-Wave testing for ENOB measurement:

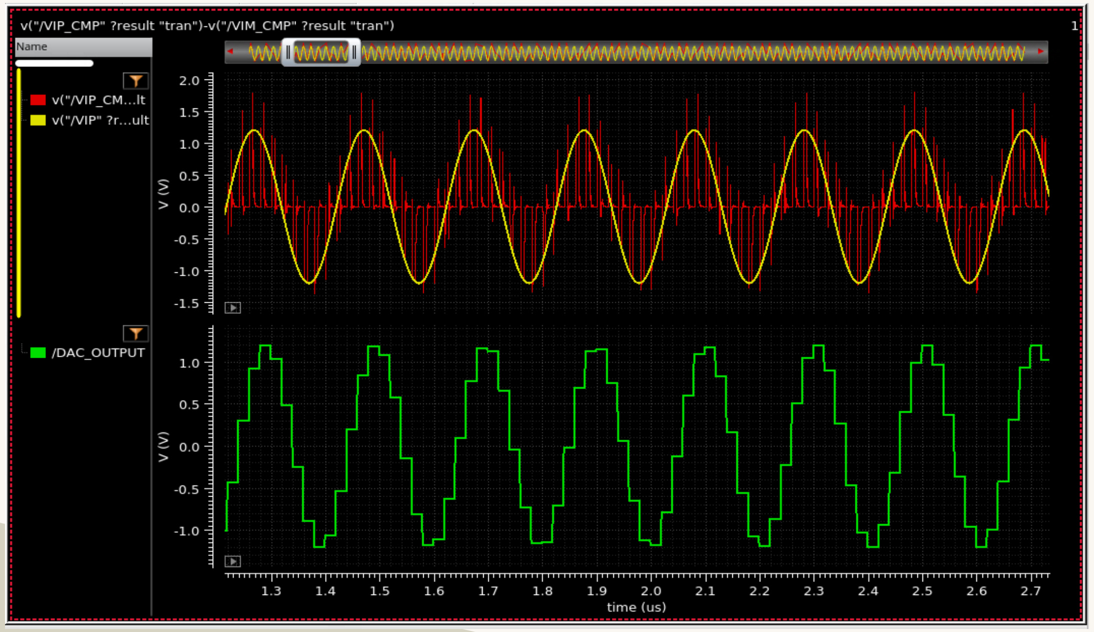
Fs = 50MHz
Fin = (m/N)*Fs

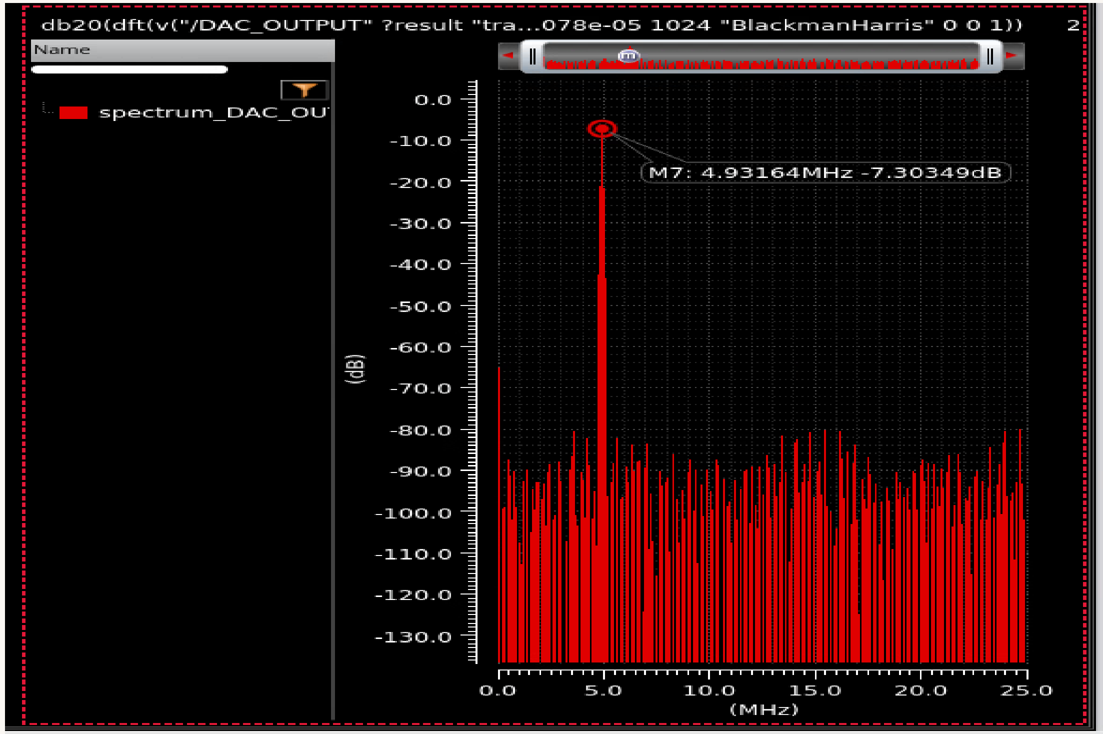
N=1024 points
Fin=4.93MHz → m=101; m, N are required to be co-prime
Fin= 19.87MHz (closer to Fs/2) → m=407; m,N are required to be co-prime.

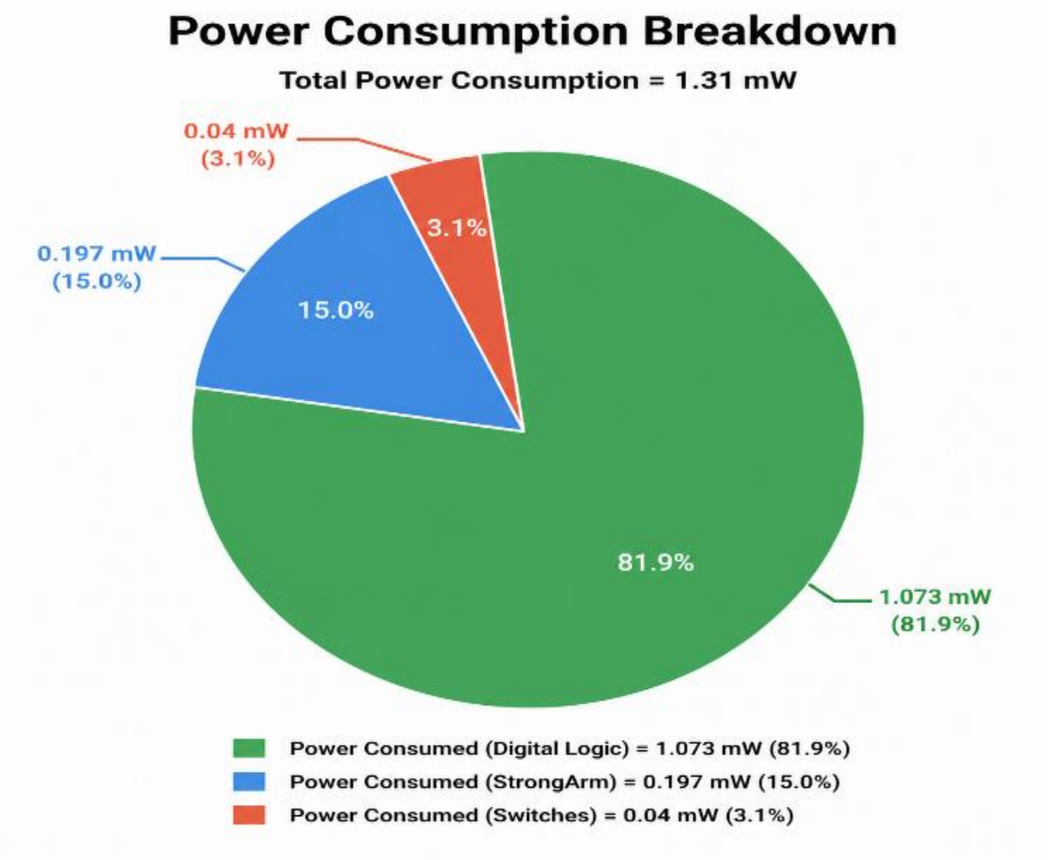
Fin=4.93MHz, Fs=50MHz, VREF=1.2 Volts, input sinusoid amplitude 600mV.
Differential Input Range : -1.2 to 1.2 Volts.

Outputs

Measurement	Value
/DAC_OUTPUT	
ENOB	9.5511909 (bits)
SINAD	59.264812 (dB)
SNR	59.264812 (dB)
SFDR	72.775474 (dBc)



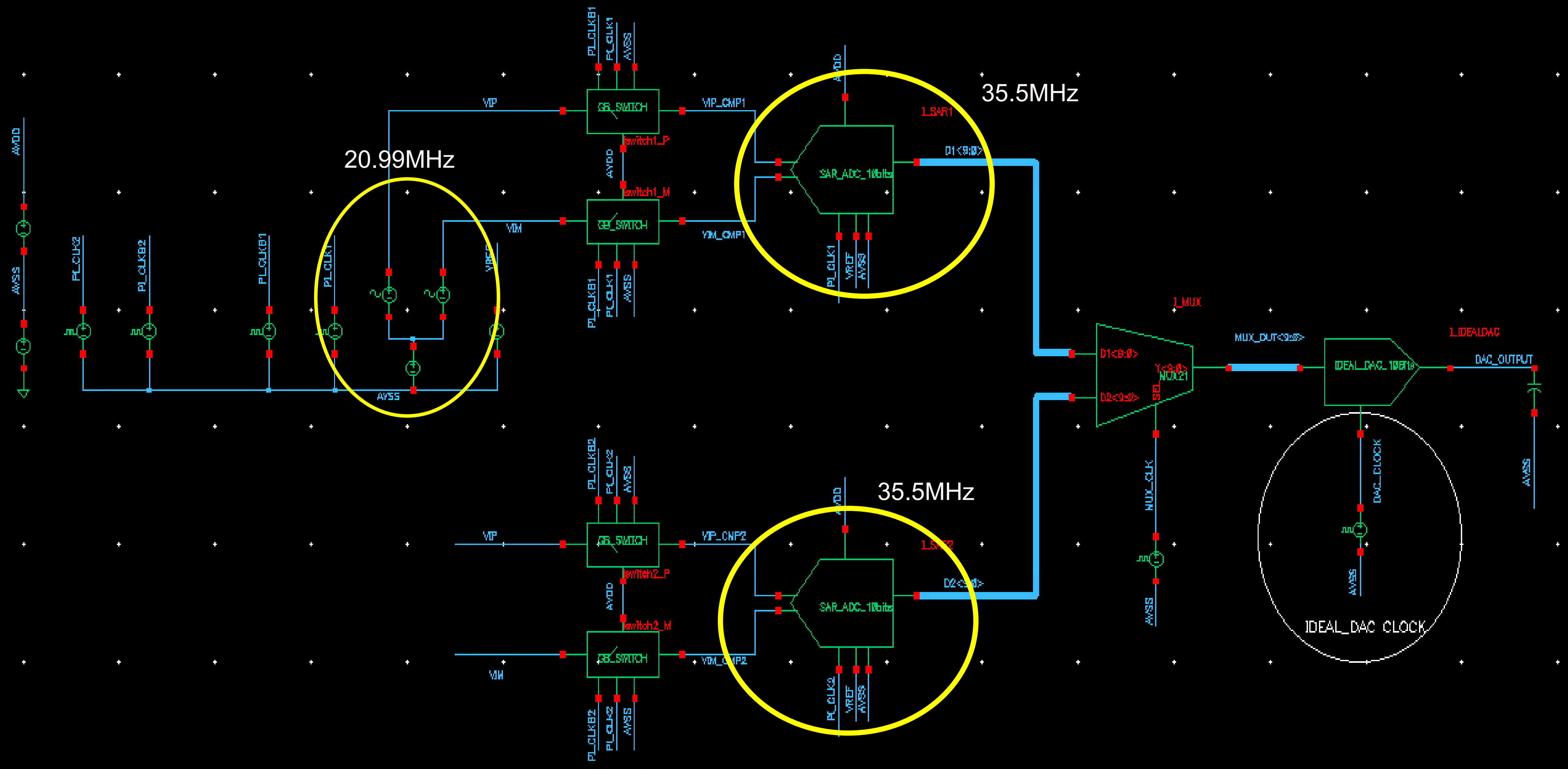




Specification	[1]	[2]	[3]	This Work
Architecture	SAR	SAR	SAR	SAR
Technology	0.13um	90nm	90nm	90nm
Supply Voltage(V)	1.2	1	1	1.2
Sampling Rate(MS/s)	50	50	40	50
Resolution	10	9	9	10
Sampling Cap(pF)	2.5	10	5	32
ENOB	9.18	7.8	8.56	9.55
Power(mW)	0.826	0.7	0.82	1.31
Schreier FOM(dB)	162	154	157.16	163
Walden FOM(fJ/conv-step)	29	65	54	33

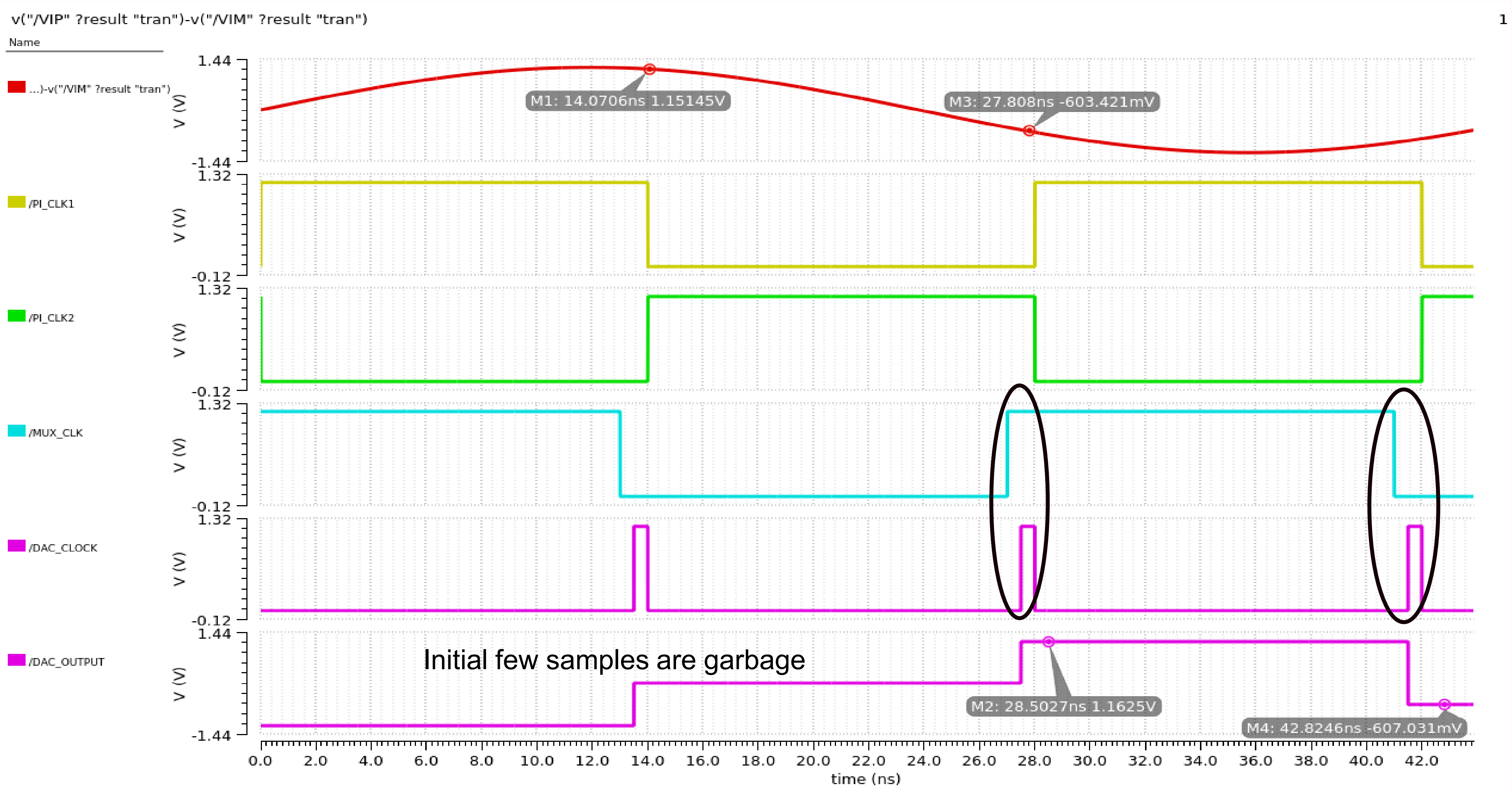
Maybe going for Dynamic DFF might help to reduce this power consumed in the digital logic and hence achieve higher Schreier FOM.

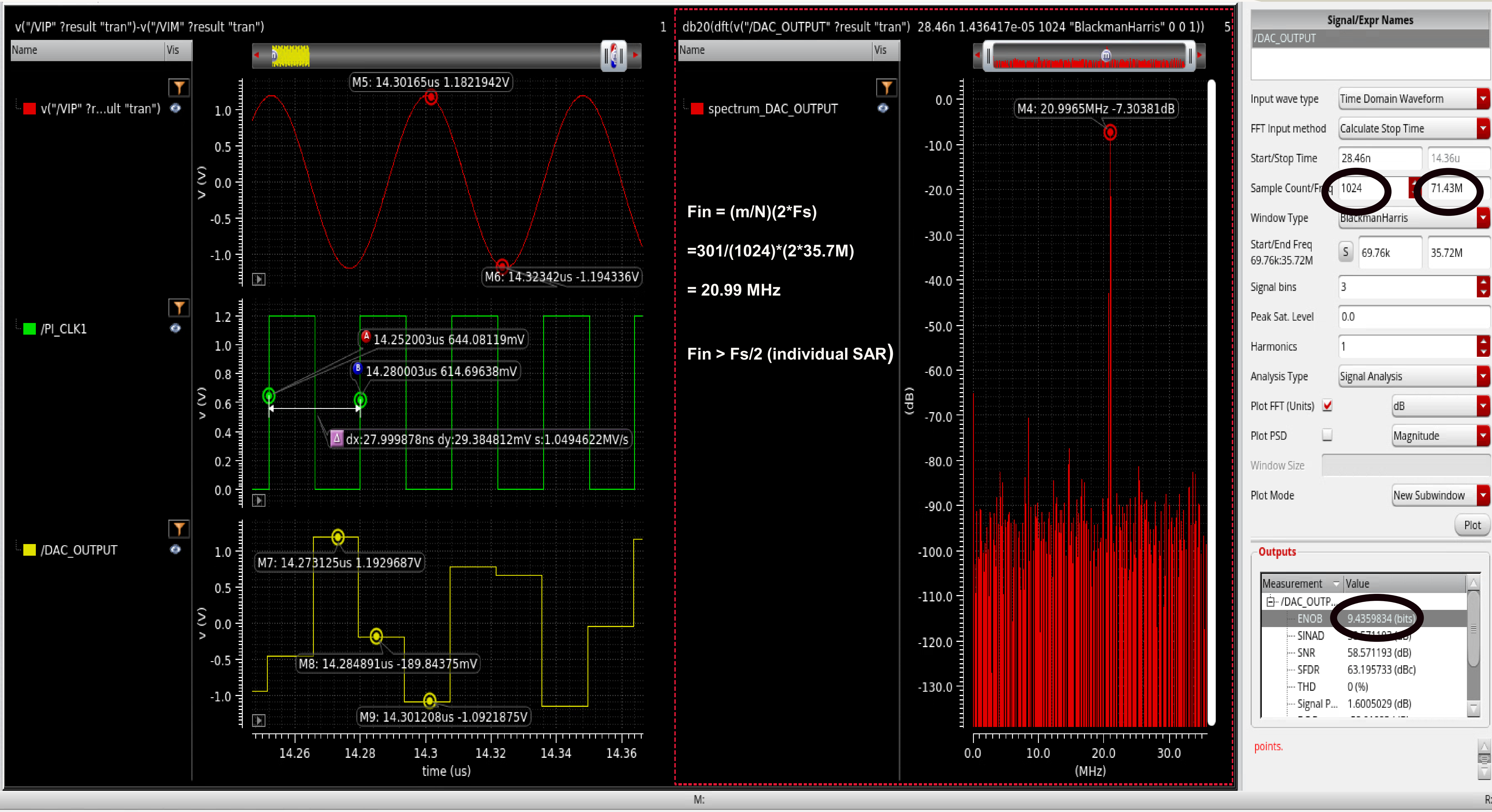
Two-Channel Time-Interleaved SAR-ADC – 71MS/s



Why is 50% duty Cycle required? – Eases the clock complexity, out of phase signals are naturally available from the oscillator

Clocking Flow





v("/VIP" ?result "tran")-v("/VIM" ?result "tran")
db20(dft(v("/DAC_OUTPUT" ?result "tran") 28.46n 1.436417e-05 1024 "BlackmanHarris" 0 0 1))
v("/VIP" ?r...ult "tran")
M5: 14.30165us 1.1821942V
M6: 14.32342us -1.194336V
/PI_CLK1
14.252003us 644.08119mV
14.280003us 614.69638mV
dx:27.999878ns dy:29.384812mV s:1.0494622MV/s
/DAC_OUTPUT
M7: 14.273125us 1.1929687V
M8: 14.284891us -189.84375mV
M9: 14.301208us -1.0921875V
time (us)
spectrum_DAC_OUTPUT
M4: 20.9965MHz -7.30381dB
Fin = (m/N)(2*Fs)
=301/(1024)*(2*35.7M)
= 20.99 MHz
Fin > Fs/2 (individual SAR)
(dB)
(MHz)
Signal/Expr Names
/DAC_OUTPUT
Input wave type Time Domain Waveform
FFT Input method Calculate Stop Time
Start/Stop Time 28.46n 14.36u
Sample Count/Freq 1024 71.43M
Window Type BlackmanHarris
Start/End Freq 69.76k:35.72M 69.76k 35.72M
Signal bins 3
Peak Sat. Level 0.0
Harmonics 1
Analysis Type Signal Analysis
Plot FFT (Units) dB
Plot PSD Magnitude
Window Size
Plot Mode New Subwindow
Plot
Outputs
Measurement Value
/DAC_OUTP...
ENOB 9.4359834 (bits)
SINAD
SNR 58.571193 (dB)
SFDR 63.195733 (dBc)
THD 0 (%)
Signal P... 1.6005029 (dB)

THANK YOU

Texas A&M University
Division of Marketing & Communication